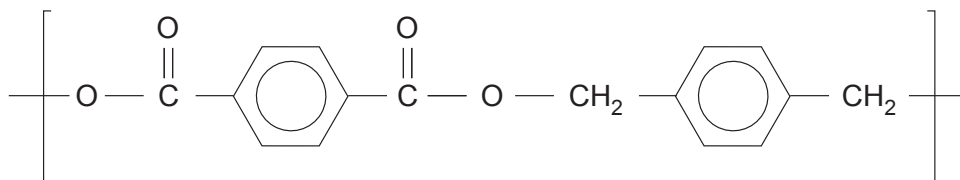
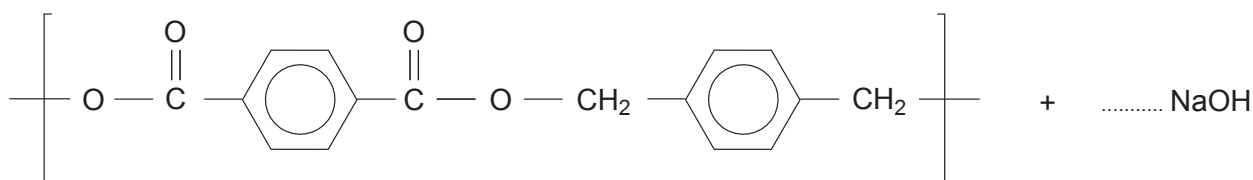


12. (a) A student was given a sample of a polyester **P**. He was told that the repeating unit was as follows.



- (i) Draw a **circle** around an ester linkage in the repeating unit. [1]
- (ii) Polyesters such as compound **P** can be hydrolysed by heating with aqueous sodium hydroxide.

Complete the balanced equation for this hydrolysis. [2]



- (iii) The formation of polyester **P** is an example of condensation polymerisation.

State how this type of polymerisation differs from addition polymerisation.
Refer to both types in your answer. [1]

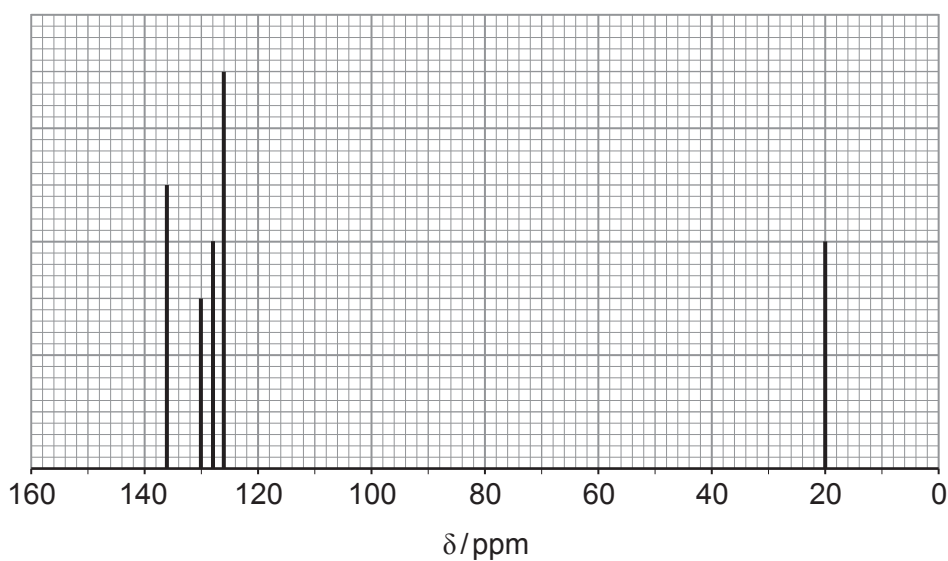
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- (iv) One of the raw materials used to make polyester **P** is methylbenzene. Under suitable conditions methylbenzene can be converted to a mixture of dimethylbenzene isomers and benzene.



Gas chromatography can be used to separate the dimethylbenzene isomers. One of the dimethylbenzene isomers has the ^{13}C NMR spectrum below.



Explain how this spectrum shows that the isomer is 1,3-dimethylbenzene.

[3]

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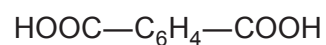
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- (v) The isomer of dimethylbenzene used to produce polyester **P** is 1,4-dimethylbenzene. In industry this isomer is oxidised by air to produce benzene-1,4-dicarboxylic acid.



During this reaction a number of other aromatic oxidation products are made. One of these has the empirical formula $\text{C}_4\text{H}_3\text{O}$.

Suggest a structure for this product giving your reasoning. State how it could be produced in the reaction between 1,4-dimethylbenzene and atmospheric oxygen.

[5]

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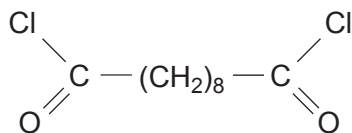
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- (b) (i) A laboratory experiment to demonstrate the formation of a polyamide uses decanedioyl dichloride.



State the name of a reagent used to produce this compound from the corresponding dicarboxylic acid, decanedioic acid. [1]

- (ii) After use, a bottle containing decanedioyl dichloride (M_r 239.1) can become contaminated with decanedioic acid.

Analysis of a 3.50 g sample of contaminated decanedioyl dichloride showed that it contained 0.977 g of chlorine by mass.

Calculate the percentage purity of this sample of decanedioyl dichloride. [2]

Percentage purity = %

- (iii) Suggest how the decanedioyl dichloride had become contaminated with decanedioic acid. [1]

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END OF PAPER

